

Approach. I treated this as a scoping problem before a coding one. The agent is a deterministic, event-sourced pipeline (ingest → profile → map → normalize → reconcile → validate → push) where every action is an append-only event carrying before/after, reason, score, and actor. The local LLM is used only where code cannot decide: breaking ties between candidate column mappings and proposing a value for an unrecognized enum. Its suggestions are re-checked against deterministic evidence, and if the model is unavailable the item escalates instead of being guessed.

How I drew the line. For every decision I asked three questions: *is it reversible*, *can code verify it*, and *what is the blast radius if it is wrong*? Whitespace, phone formats, exact-key duplicates, and a date column where one day exceeds 12 pass all three, so the agent acts. A column that could be employee_id or manager_id, dates where every day is ≤ 12, a salary that differs across systems, or two near-identical people with the same DOB fail at least one, so a human decides once and the answer is remembered as a rule until disabled. On the fixtures that took review cards from 7 to 3, and the remaining ones are record-level decisions that cannot be generalized.

1. ESCALATION BOUNDARY DEFENSE MATRIX (MIRRORING BRIEF CRITERIA 2 & 3)

Trigger / Scenario	Agent Action	Escalate?	Rationale & Boundary Defense
Column mapping: ambiguous vs high-margin	Auto-maps if conf ≥ 0.75 & margin ≥ 0.20; calls LLM & prompts if margin < 0.20 (ref)	< 0.20	High-margin maps autonomously; margin < 0.20 (e.g. emp_id vs mgr_id) destroys reporting hierarchy if guessed.
Sensitive field mapping (Salary / DOB)	Batches candidate mappings into 1 confirmation card	Yes	High blast-radius compliance fields; one-click approval, remembered as rule until disabled.
Uncleanable value (Enum "LOA")	Proposes closest canonical status ("On Leave")	Yes	Unrecognized domain enum cannot be guessed; batched into 1 card, saved as rule until disabled.
Ambiguous dates vs consensus	Prompts once if all days ≤ 12; auto-resolves if any day > 12	≤ 12	04/05/2021 is ambiguous; guessing shifts hire dates. If any day > 12, only one format parses every value in the column.
Near-duplicate identity	Fuzzy name match ≥ 90 + identical DOB	Yes	Merging two distinct people into one ID is irreversible data loss. Surfaces side-by-side diff.
Sensitive conflict (Salary ₹80k vs ₹88k)	Withholds auto-merge; surfaces side-by-side diff	Yes	Cross-source payroll discrepancy requires human sign-off; non-sensitive conflicts use source precedence.
Record fails validation twice	Validates → Auto-fixes → Re-validates → Quarantines	Yes	Missing required PII (e.g. email) cannot be synthesized; quarantined immediately for review.
Whitespace, casing, phone & exact duplicates	Strips blanks, title-cases, normalizes E.164 (+91); merges exact matches	No	Reversible; deterministic regex verification; non-conflicting null fills; zero information loss.
Target API: 503 transient vs 422/400 error	503 retries with backoff & jitter; 422/400 parks record & halts retry	4xx Only	Network blips succeed on retry; malformed payloads cannot succeed and require human correction/assignment.

2. DELTA SOLUTIONING & PRODUCTION ARCHITECTURE

- **Resolution Memory (Learned Rules Engine)**
- Human resolutions (mapping, enums, date format) are fingerprinted into client-scoped rules table.
- On re-runs, learned rules fire *before* the escalation gate, auto-resolving repeat ambiguities.
- **Pre-push review cards drop from 7 → 3 (-57%)** on Run 2 on the planted fixture set.

- **Audit Trail, Idempotency & Rollback**
- **Append-Only Event Spine:** SQLite triggers prevent UPDATE/DELETE on events for auditability.
- **Idempotency & Replay:** Target push uses SHA256 Idempotency-Key; offline LLM cache uses SHA256 prompt hashing.
- **Compensating Rollback:** Outages trigger automated rollback (DELETE /stub/employees/{id}) to preserve target purity.

3. EMPIRICAL VERIFICATION & PRIVACY ARCHITECTURE

SOURCE → MERGED:	52 → 29	PRE-PUSH REVIEW CARDS:	7 → 3 (-57%)	ZERO-TOUCH EMPLOYEES (RUN 2):	26 of 29 (90%)*	ACTION AUTONOMY SCORE:	88.5% → 95.1%
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*Counting rule: In Run 2, 3 review cards touch 3 records (salary conflict affects 1 employee, unresolvable validation failure affects 1 employee, near-duplicate card involves 2 candidates counted as 1 distinct unresolved employee slot). 26 of 29 canonical records require zero human touch. Autonomy score = (Automated Decisions / Total Pipeline Actions) × 100%.

Enterprise Perimeter Isolation: The hosted demo runs strictly on synthetic test data with an evaluator “Reset Demo” control. In production enterprise deployments, local LLM inference (e.g. Ollama / on-prem vLLM) executes entirely inside the client VPC boundary with **no row-level data leaving the client VPC**. Cross-tenant synonym matching is strictly opt-in and operates only on anonymized header tokens, never row-level PII.

4. PRODUCTION ROADMAP: WHAT I WOULD BUILD NEXT

Topological Dependency Graph: Sequence target entity pushes (Departments → Cost Centers → Managers → Employees) to eliminate referential integrity rejections.

Cross-Tenant Anonymized Embeddings: Cluster anonymized column header patterns across clients to suggest high-probability synonym maps as pre-warmed recommendations while preserving strict tenant isolation.

Automated Schema Evolution: Detect when client source exports drift (new columns or shifted formats) and generate automated diff migration proposals.